

Total No. of Questions : 8]

SEAT No. :

PC1773

[Total No. of Pages : 3

[6353]-92

T.E. (Electronics & Telecommunication Engineering)

DIGITAL COMMUNICATION

(2019 Pattern) (Semester - I) (304181)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is QAM? Draw and Explain Block diagram of generation and reception of the QAM. [8]

b) Calculate bandwidth requirement and minimum separation (Euclidean distance) of Signal points in signal space for

i) 16-PSK

ii) 16-FSK

iii) 16-QAM. Given that input bit rate is 8kbps and bit energy is $1 \times 10^{-4} \text{J}$. [9]

OR

Q2) a) What is M-ary FSK? Draw and Explain Block diagram of generation and reception of the M-ary FSK. [8]

b) What is OFDM? Draw and Explain block diagram of generation and reception of the OFDM. [9]

Q3) a) What is PN sequence? Explain properties of PN sequence. [9]

b) A DS-SS BPSK system has $f_b = 3 \text{kbps}$, $N_0 = 10^{-10} \text{W/Hz}$ and is receiving signals with $P_e = 10^{-7}$ in the presence of single tone jammer whose received power is ten times larger than original signal. Calculate the jamming margin. Assume $Q(5.2) = 10^{-7}$. [9]

OR

P.T.O.

- Q4) a)** Write a short note on following. [9]
- i) Slow FHSS
 - ii) Fast FHSS

- b) Explain DSSS BPSK system in detail. [9]

- Q5) a)** Explain binary symmetric channel in detail and find capacity of channel. [8]

- b) Apply Shannon-Fano code for following message ensemble and find coding efficiency

$$X = [x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8]$$

$$P = [1/4 \ 1/8 \ 1/16 \ 1/16 \ 1/16 \ 1/16 \ 1/8 \ 1/8].$$
 [9]

OR

- Q6) a)** A discrete source transmits messages x_1 and x_2 with probability $3/4$ and $1/4$. The source connected to the binary symmetrical Channel with $p(y_1/x_1) = 2/3$. Calculate all entropies and mutual information. [8]

- b) Apply Huffman coding for the following message ensemble.

$$X = [x_1 x_2 x_3 x_4 x_5 x_6 x_7]$$

$$P = [0.45 \ 0.15 \ 0.1 \ 0.1 \ 0.08 \ 0.08 \ 0.04] \text{ and find coding efficiency.}$$
 [9]

- Q7) a)** For a systematic linear block code, the three parity check digits, are given by

$$C_4 = d_1 \oplus d_2 \oplus d_3$$

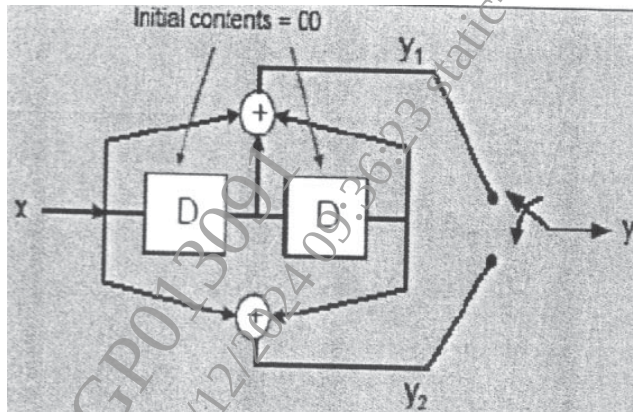
$$C_5 = d_1 \oplus d_2$$

$$C_6 = d_1 \oplus d_3$$

[9]

- i) Construct Generator Matrix
- ii) Construct All Code generated by this matrix
- iii) Determine error correcting capability
- iv) Prepare suitable decoding table
- v) Decode the received words 0 0 0 1 1 0

b)



For given convolutional encoder draw three graphical representation.[9]

OR

Q8) a) Draw the encoder and syndrome calculator for the generator polynomial $g(x) = 1 + x^2 + x^3$ and obtain the syndrome for the received codeword 1 0 0 1 0 1 1. [9]

b) Explain properties of Linear Block code and Cyclic Code with example. [9]

* * *